IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 4 February 2026, accepted 26 February 2026, date of publication 4 March 2026, date of current version 10 March 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3670441

SURVEY

Accelerating Pharmaceutical Innovation Through AI: A Systematic Review of Machine Learning and Deep Learning Technologies for Drug-Drug Interaction Prediction

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ABSTRACT Drug–Drug Interactions (DDIs) pose a significant challenge to patient safety and therapeutic efficacy. Traditional experimental methods for detecting DDIs are labor-intensive, costly, and cannot scale to the vast number of potential drug combinations. To address these challenges, Artificial Intelligence (AI) approaches—particularly Machine Learning (ML) and Deep Learning (DL)—have emerged as effective computational tools for predicting DDIs and supporting clinical decision-making. This study systematically reviews the application of ML and DL methods to DDI prediction based on the available English-language literature, evaluating their contributions, methodological strengths, limitations, and future opportunities. Following PRISMA guidelines, a Systematic Literature Review (SLR) of 75 primary studies was conducted, analyzing commonly used datasets, model architectures, algorithms, and evaluation metrics, with an emphasis on clinical relevance and potential to prevent adverse drug events. Similarity-based, network-based, and deep learning approaches dominate the field. While DL models often achieve higher predictive accuracy, they face challenges such as overfitting and limited interpretability, reducing their clinical trustworthiness and adoption in real-world settings. AI-driven methods have significantly advanced DDI prediction and demonstrate strong potential to enhance drug safety monitoring and clinical decision support. However, critical gaps remain, particularly in applying advanced AI paradigms such as transfer learning and multi-agent systems. Future research should prioritize larger, more balanced datasets and improved interpretability to ensure reliable, clinically actionable outcomes and foster wider adoption in healthcare practice.

INDEX TERMS Adverse drug events (ADE), AI in drug discovery, computational pharmacology, drug-drug interactions.

I. INTRODUCTION

Drug-drug interactions (DDIs) play a crucial role in determining the safety and effectiveness of medical treatments. Particularly in cases where multiple drugs are administered concurrently, accurately predicting these interactions is essential to prevent potential adverse effects. The increasing global aging population and the widespread use of polypharmacy necessitate the thorough investigation of such interactions. For instance, antagonistic interactions may negatively impact the pharmacokinetic or pharmacodynamic properties of drugs, potentially leading to severe health complications.

The advent of Artificial Intelligence (AI) has significantly transformed the landscape of pharmaceutical research and drug discovery. Unlike conventional trial-and-error methods that are often costly, time-intensive, and statistically restrictive, AI-driven systems offer adaptive, data-intensive approaches that can extract meaningful patterns from complex biomedical data. AI facilitates the identification of drug candidates, prediction of adverse effects, optimization of molecular structures, and the acceleration of clinical

The associate editor coordinating the review of this manuscript and approving it for publication was Vincenzo Conti.

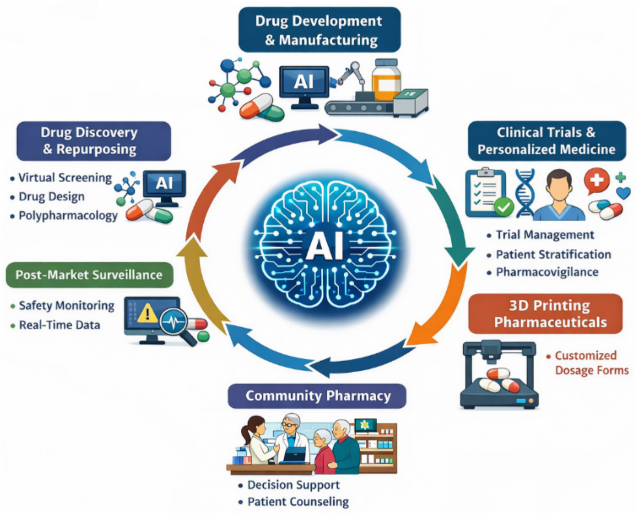

FIGURE 1. The application areas of artificial intelligence in the pharmaceutical field, ranging from drug design to community pharmacy.

decision-making processes. These advantages not only improve the precision of drug discovery pipelines but also substantially reduce the time and financial investments required for novel drug development. As illustrated in Figure 1, the integration of Artificial Intelligence (AI) covers a broad spectrum of the pharmaceutical lifecycle. In the early stages, AI accelerates drug screening, drug design, and drug repurposing, enabling the identification of potent candidates while managing complex polypharmacology scenarios. Moving into production, AI-driven systems optimize pharmaceutical product development, manufacturing, and product management, ensuring efficiency and high-quality standards. During clinical and post-market phases, these technologies enhance pharmacovigilance, clinical trial management, and personalized medicine, allowing for the real-time monitoring of safety signals. Furthermore, innovative applications such as 3-D printing of pharmaceuticals provide tailored dosage forms. Finally, AI supports community pharmacy by providing pharmacists with advanced decision support tools to improve patient care and safety at the point of dispensing.

Although DDI research has traditionally relied on laboratory experiments, these methods are both costly and time-consuming. Furthermore, it is impractical to experimentally test all possible drug combinations. In this context, in silico approaches such as computational modeling, molecular simulations, and bioinformatics analyses have gained significant attention in recent years. Among these, Machine Learning (ML) and Deep Learning (DL) techniques have emerged as particularly powerful tools for predicting drug interactions. Such techniques can process extensive datasets, uncover intricate relationships, and offer deeper insights into molecular mechanisms.

Various approaches, including similarity-based, network-based, and machine learning-based methods, have been proposed in the literature for DDI prediction. Similarity-based methods rely on similarity measures between drug profiles, whereas network-based approaches focus on drug targets and protein-protein interaction networks. Machine learning-based methods, on the other hand, often integrate heterogeneous data sources to develop high-performance predictive models. Nevertheless, these approaches often face challenges including overfitting and limited model interpretability [1].

The objective of this study is to systematically examine the ML- and DL-based methods used for drug interaction prediction, identify the strengths and weaknesses of existing approaches, and highlight research gaps in this field. The study evaluates current methodologies in terms of both model performance and biological interpretability.

To provide a comprehensive overview of the current state of DDI prediction methods, a Systematic Literature Review (SLR) was conducted. This study differs from other literature reviews in that it employs a systematic search strategy across electronic databases, selects relevant studies in a structured manner, extracts necessary data, and synthesizes the findings to answer predefined research questions. The SLR process involves reviewing primary studies published on the topic across various electronic databases. To achieve this, a set of research questions was formulated, and relevant information from the literature was collected to address these questions explicitly. Additionally, the findings were analyzed to identify areas requiring further investigation.

The remainder of this paper is structured as follows: Section II outlines the foundational concepts related to drug–drug interaction prediction and provides an overview of pertinent literature, thereby establishing the rationale for the present study. Section III delineates the research methodology, detailing the formulation of research questions, study design, and analytical procedures employed. Section IV presents the empirical findings derived from the analysis. Section V offers a comprehensive discussion of the results in light of the research objectives and existing literature. Section VI examines the ethical considerations associated with the study. Finally, Section VII concludes the paper by summarizing the main contributions and proposing potential directions for future research.

II. BACKGROUND AND RELATED WORK

A. DRUG-DRUG INTERACTIONS: OVERVIEW AND CHALLENGES

Drug-drug interactions (DDIs) occur when one drug alters the pharmacokinetic or pharmacodynamic properties of another, potentially leading to undesirable side effects or toxic effects. This phenomenon is particularly critical for elderly populations and individuals with chronic diseases. Most adverse effects arise from interactions involving enzymes and transporter proteins that play a key role in drug metabolism [2], [3].

Traditionally, the detection of drug interactions has been conducted through in vitro and in vivo studies; however, these methods are time-consuming and costly, thus limiting their applicability. Moreover, it is impractical to test all possible drug combinations in clinical trials. These limitations have driven the development of data-driven approaches for predicting drug interactions.

B. COMPUTATIONAL METHODS FOR DDI PREDICTION

In recent years, these in silico methods have been increasingly preferred for predicting drug-drug interactions (DDIs). These approaches enable the molecular-level understanding of DDIs by analyzing large datasets. The literature highlights three primary approaches for DDI prediction: similarity-based, network-based, and machine learning-based methods [4].

Similarity-based methods predict interactions by evaluating similarities between drug profiles. Various data sources, such as drug structural profiles, gene expression data, and phenotypic profiles, are utilized. However, these methods can be sensitive to noise, and the reliability of similarity measures may be limited.

Network-based methods predict interactions based on the proximity of genes targeted by two drugs, leveraging protein-protein interaction networks or drug similarity networks. While these approaches enhance the understanding of biological mechanisms, they may encounter challenges such as missing data and network noise. Recent advances such as the DRAW+ framework, which incorporates noise filtering via Singular Value Decomposition and attention-based random walks, have demonstrated state-of-the-art performance in drug–disease association prediction [5].

Machine learning and deep learning approaches aim to integrate heterogeneous data sources to develop high-performance predictive models. Although these methods yield more accurate results compared to traditional approaches, they also face challenges such as overfitting and model interpretability [6].

C. DATA RESOURCES FOR PREDICTING DRUG INTERACTIONS AND SIDE EFFECTS

In recent years, machine learning-based approaches have demonstrated considerable potential for predicting adverse effects caused by pharmaceutical substances. These approaches rely heavily on diverse and comprehensive datasets that include detailed information about the chemical structures of drugs, their biological targets such as proteins, and previously observed side effects. Such datasets are essential for identifying and characterizing possible adverse effects and the properties of molecular targets. Table 1 provides an overview of the major datasets that are currently used or considered useful in research related to predicting harmful effects of drugs. This table is structured to provide a multi-dimensional view of each resource, categorized by latest update versions, primary interaction types (e.g., chemical-protein, network-based, or mechanistic associations), and systematic data descriptions. To ensure that the survey reflects contemporary advancements, high-impact datasets with 2024 and 2025 updates, including DrugBank, OpenFDA, and the Comparative Toxicogenomics Database (CTD), have been integrated. Additionally, for dynamic

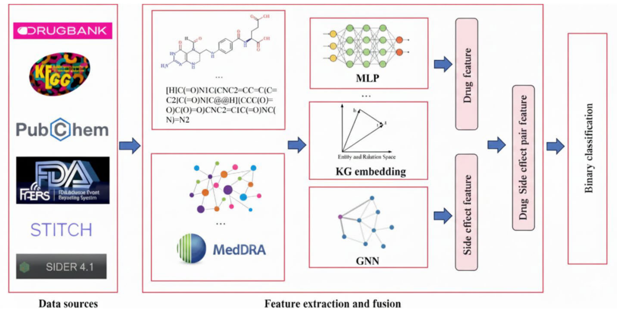

FIGURE 2. Machine learning-based pipeline for drug–side effect association prediction.

resources subject to quarterly update cycles—such as PubChem and FAERS—the most recent iterations available as of early 2025 are incorporated to maintain temporal relevance. These resources contain valuable evidence regarding harmful drug effects, inter-drug interactions, and bio-molecular features essential for target protein analysis. By providing a rich and structured source of data, these resources support the development, training, and evaluation of predictive models and play a fundamental role in improving the accuracy and reliability of machine learning approaches in this field.

The data flow and feature extraction process involved in predicting drug–side effect associations using machine learning models is illustrated in Figure 2. As depicted in the computational pipeline, the process commences with the aggregation of heterogeneous data from diverse sources including DrugBank, PubChem, and FAERS. These raw data undergo feature extraction where drug chemical structures and side-effect profiles are processed through deep learning architectures, such as multi-layer perceptrons (MLP) and graph neural networks (GNN), to generate high-dimensional embeddings. Subsequently, these fused drug-side effect representations are utilized by a binary classification layer to predict potential adverse associations [7].

D. ADVANCING TECHNIQUES AND APPROACHES

Multimodal Deep Learning: This approach enables more detailed and accurate predictions by utilizing four types of drug features (chemical substructures, targets, enzymes, and pathways). These models provide faster and more cost-effective results compared to laboratory-based tests [8].
SMILES-Based Models: Utilizing SMILES strings—which represent drug chemical structures—has enhanced the extraction of molecular features and improved the accuracy of DDI prediction [28].
Hybrid Models: The combination of methods such as Naive Bayes, Random Forest, and XGBoost has led to significant improvements in prediction performance. Techniques like SMOTE have been particularly applied to reduce data imbalances [29].

Beyond multimodal and hybrid frameworks, recent advances have introduced **multi-view and multi-channel attention-based models** for DDI prediction. These approaches

TABLE 1. Major databases used for detecting drug side effects and interactions.

Database	Latest Update	Type	Interaction / Data Description
SIDER [8]	2015/10 V4.1	Adverse Reactions	A database containing information on marketed medicines and their recorded adverse reactions.
STITCH [9]	2016/01 V5	Chemical-Protein	A database of known and predicted interactions between chemicals and proteins.
BIOSNAP [10]	2018	Network	A biological network collecting various types of interactions between FDA-approved drugs.
TWOSIDES [11]	2019/11	Safety Signals	A database of DDIs safety signals mined from the FDA's Adverse Event Reporting System.
Offsides [12]	2019/11	Side Effects	A database of individual drug side effect signals mined from the FDA's Adverse Event Reporting System.
DDInter [13]	2020/09 V1.0	DDI Association	A comprehensive and open-access database providing abundant annotations for each DDI association.
DailyMed [14]	2021/10	Drug Labeling	A database containing labeling submitted to the Food and Drug Administration (FDA) by companies.
MecDDI [15]	2023/02	Mechanistic	A database offering the mechanism underlying 110,000 DDIs by explicit description and graphic illustration.
ADReCS [16]	2023/03 V3.2	Ontology	A comprehensive drug side effect ontology database providing both standardization and hierarchical classification.
ChEMBL [17]	2023/05 V33	Bioactive	A manually curated database of bioactive molecules with drug-like properties including chemical and bioactivity data.
TTD [18]	2023/07	Therapeutic	A database providing information about therapeutic targets and corresponding drugs information.
KEGG [19]	2023/09 V108	Pathways	A database resource for understanding high-level functions and utilities of the biological system.
PubChem [20]	Quarterly Update	Chemical	A database containing information on physical properties and biological activities of compounds.
PharmGKB [21]	Quarterly Update	Pharmacogenomics	A comprehensive resource that curates knowledge about the impact of genetic variation on drug response.
Drugs [22]	Quarterly Update	Clinical Info	Accurate and independent information on more than 24,000 prescription drugs and over-the-counter medicines.
UniProt [23]	Quarterly Update	Protein Data	A database providing protein sequence and functional information.
FAERS [24]	Quarterly Update	Adverse Event	A database containing information on adverse event reports, medication error reports, and product quality complaints.
DrugBank [25]	2024/11 (actively maintained)	Drug–Drug Interaction	A comprehensive, manually curated database containing drug–drug interactions, drug targets, enzymes, transporters, and detailed pharmacological information.
OpenFDA [26]	2024 (continuous update, accessed 2025)	Adverse Events	An open-access platform providing continuously updated adverse event data derived from the FDA Adverse Event Reporting System (FAERS) for large-scale drug safety analysis.
CTD [27]	2024/09 (actively maintained)	Chemical–Gene–Disease	A curated resource integrating chemical–gene, chemical–disease, and gene–disease associations to facilitate mechanistic studies of drug safety, toxicity, and related biological processes.

DDI = drug–drug interaction, FAERS = FDA adverse event reporting system, FDA = U.S. Food and Drug Administration, KEGG = Kyoto encyclopedia of genes and genomes.

integrate drug attributes (such as chemical substructures, targets, enzymes, and pathways) with drug-related entities derived from knowledge graphs into a unified representation. By employing a multichannel attention mechanism, the models can adaptively assign importance weights to different feature views, which has been shown to enhance predictive performance over several existing state-of-the-art methods [30].

E. MACHINE LEARNING TECHNIQUES FOR DDI PREDICTION

Machine Learning plays a significant role, particularly in the analysis of drug profiles and the prediction of interactions. In the literature, models based on Support Vector Machines, Decision Trees, Random Forests, and Deep Learning are widely used. Deep Learning-based models typically include Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and their hybrid architectures. Notably, algorithms such as Long Short-Term Memory (LSTM) stand out for modeling time series data [31], [32], [33].

As summarized in Table 2, traditional machine learning-based methods for DDI prediction can be broadly grouped into similarity-based, feature vector-based, and semi-supervised approaches. Similarity-based methods such as SIde-effect and TArget based Regularization (SITAR), K-Nearest Neighbors (KNN), Kronecker Regularized Least

TABLE 2. **Traditional machine learning-based methods for drug–target interaction prediction.**

Reference	Year	Model	Type	Metrics
Xia et al. [34]	2010	NetLapRLS	Semi-supervised	Sensitivity, Specificity
Perlman et al. [35]	2011	SITAR	Similarity-based	AUC, AUPR
Laarhoven et al. [36]	2011	RLS-Kron	Semi-supervised	AUC, AUPR
Yu et al. [40]	2012	Random Forest	Feature vector-based	SE, SP, CO, AUC
Wang and Zeng [38]	2013	RBMs	Semi-supervised	AUC, AUPR
Shi et al. [39]	2015	KNN	Similarity-based	AUC, AUPR
Pahikkala et al. [40]	2015	KronRLS	Similarity-based	CI, AUC
Faulon et al. [41]	2016	SVM	Feature vector-based	ROC
Wang et al. [42]	2016	Adaboost-SVM	Feature vector-based	EF, ROC
He et al. [43]	2017	SimBoost	Similarity-based	MSE, CI
Buza and Peska [44]	2017	BLM & KNN	Similarity-based	AUC, AUPR
Olayan et al. [45]	2018	Random Forest	Similarity-based	AUC, AUPR
Aghakhani et al. [46]	2018	K-Means	Semi-supervised	Precision, Recall, AUC
Ezzat et al. [47]	2019	Random Forest	Feature vector-based	AUPR

AUC = Area Under the Receiver Operating Characteristic Curve, AUPR = Area Under the Precision-Recall Curve, CI = Concordance Index, MSE = Mean Squared Error, SVM = Support Vector Machine, RF = Random Forest, RLS = Regularized Least Squares, KNN = K-Nearest Neighbors, SE = Sensitivity, SP = Specificity, CO = Coverage, EF = Enrichment Factor, ROC = Receiver Operating Characteristic.

Squares (KronRLS), and Similarity-Based Gradient Boosting (SimBoost) rely on drug–drug or drug–target similarity metrics. These methods are relatively simple and interpretable, yet they often face limitations when dealing with scalability and novel drugs. Feature vector-based methods including Random Forest (RF), Support Vector Machine (SVM), and Adaptive Boosting with Support Vector Machine (Adaboost-SVM) encode drugs and targets as numerical descriptors, providing flexibility but depending heavily on feature engineering. Semi-supervised methods, such as Network-based Laplacian Regularized Least Squares (NetLapRLS), Regularized Least Squares with Kronecker Product Kernel (RLS-Kron), and Restricted Boltzmann Machines (RBMs), combine labeled and unlabeled data, thus improving generalization under sparse data conditions, although they are often sensitive to parameter tuning and class imbalance.

Artificial intelligence plays an increasingly central role in modern drug discovery and drug–drug interaction prediction, integrating both traditional and deep learning methodologies within a unified computational framework. The unified computational framework, illustrated in Figure 3, operationalizes the integration of multi-source pharmaceutical data with advanced AI methodologies. The process begins with Data Acquisition from heterogeneous resources such as DrugBank, PubChem, and SIDER. These raw inputs are processed through a Methodological Spectrum that encompasses both traditional algorithms (e.g., Random Forest, SVM) and deep learning architectures (e.g., CNN, RNN, DNN). As shown in the figure, this integrated pipeline supports a wide array of Pharmaceutical Applications, including target identification via AlphaFold, molecular design using DeepDelta, and the evaluation of ADMET profiles through models like Deep-PK. Crucially, for drug-drug interaction (DDI) prediction, the framework utilizes cutting-edge models such as MUFFIN and Drug GPT, effectively bridging the gap between raw data and clinical safety insights.

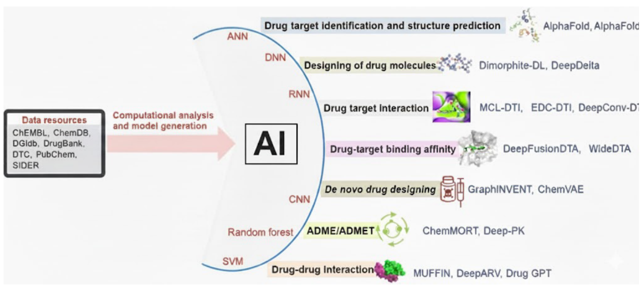

FIGURE 3. **Role of artificial intelligence in drug discovery and drug–drug interaction prediction.**

F. DEEP LEARNING TECHNIQUES FOR PREDICTING DRUG-DRUG INTERACTIONS

Machine Learning plays a significant role in analyzing drug profiles and predicting drug interactions. However, in recent years, Deep Learning methods have gained more attention due to their ability to process complex and high-dimensional biomedical data. Among these, Convolutional Neural Networks (CNNs) are the most widely applied, effectively capturing chemical structures and biological properties of drugs, especially when working with visual or sequential molecular representations. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks have also been employed, particularly in studies involving temporal or sequential pharmacological data, although their use is less common compared to CNNs. In addition, Graph Neural Networks (GNNs) have become increasingly prominent, as they are well-suited for modeling drugs and targets in the form of molecular graphs. More recent approaches also integrate 3D structural information (e.g., 3D CNN, Uni-Mol) to improve predictive accuracy by incorporating spatial molecular interactions [48], [49].

However, the field has undergone an enormous evolution recently, shifting from these "classic" architectures toward Transformer-based models and Large Language Models (LLMs). Transformer architectures, such as GraphormerDTI and FusionDTI, utilize self-attention mechanisms to capture long-range dependencies within molecular sequences and graphs more effectively than traditional CNNs or RNNs. Furthermore, the integration of cross-attention mechanisms (e.g., CAT-DTI) and heterogeneous deep attention networks (e.g., DrugMAN) has enabled more sophisticated modeling of the interactive space between drugs and proteins. Most recent advancements in 2024 and 2025, such as GS-DTI, leverage pre-trained protein language models (Pre-trained Protein Language Models) and multi-modal fusion strategies, marking a transition toward "foundation models" in drug-target interaction (DTI) prediction.

As summarized in Table 3, while CNN- and GNN-based models dominated early studies, the latest research increasingly adopts Transformer-based and hybrid Attention-GNN architectures. These models, along with 3D space–based methods (e.g., Uni-Mol, HiGraphDTI), extend predictive capability beyond traditional feature learning, with evaluation metrics such as MSE, CI, AUROC, and F1-score being widely reported to benchmark their superior performance.

III. RESEARCH METHODOLOGY

This study aims to conduct a systematic literature review to examine drug-drug interactions. The research focuses on analyzing the use of Deep Learning and Machine Learning methods for drug interaction prediction and analysis within a specific period.

A. REVIEW PROTOCOL

The scope and objectives of the study were clearly established at the outset to define the direction and boundaries of the systematic review. Based on these objectives, detailed inclusion and exclusion criteria were formulated to ensure that only relevant and high-quality studies were considered. The data collection process was carried out using both primary scientific databases and supplementary information sources. In particular, 120 records were retrieved through database searches from platforms such as PubMed, DrugBank, Web of Science, Scopus, ScienceDirect, IEEE Xplore, ACM Digital Library, and Wiley. To ensure comprehensiveness, an additional 50 records were identified through other sources, including Google Scholar, ClinicalTrials.gov, organizational reports, and citation tracking.

All retrieved records were compiled into a single database, after which 20 duplicate entries were removed to prevent redundancy. During the initial screening phase, 100 records were examined based on their titles and abstracts, leading to the exclusion of 30 studies that were considered out of scope or irrelevant to the study's objectives. The remaining 70 studies proceeded to full-text screening, where 10 reports could not be retrieved. Following full-text review, 15 additional studies were excluded due to reasons such as lack of dataset information, poor methodological design, or irrelevance to machine learning or deep learning approaches, resulting in 60 studies deemed eligible from the database search pathway.

Simultaneously, the studies identified from other sources (n = 50) underwent a similar evaluation process. 10 of these reports could not be retrieved, and 5 were excluded after full-text assessment due to inadequate methodological quality or insufficient dataset details, leaving 35 eligible studies from alternative sources.

In total, 75 studies that met all predefined inclusion criteria and demonstrated sufficient methodological rigor were included in the final synthesis.This multi-phase screening ensured that only studies providing robust and scientifically relevant findings were incorporated into the review. Two reviewers (Z.B. and T.T.B.) independently screened the titles, abstracts, and full texts to minimize selection bias. Discrepancies were first addressed through a structured consensus-seeking discussion between the primary reviewers, where each study's eligibility was re-evaluated against the predefined Quality Score (QS1–QS5) criteria. In cases where a consensus could not be reached, the second author (T.T.B.) acted as a senior arbitrator to make the final determination based on the study's alignment with the research objectives and methodological rigor.

Data extraction was performed using a standardized form capturing study characteristics (authors, year, dataset, model type, metrics, and key findings). Data were cross-checked by two reviewers to ensure consistency.

To maintain methodological transparency and reproducibility, the study selection process adhered strictly to the PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. The workflow—covering identification, screening, eligibility assessment, and final inclusion—is summarized in Figure 4, which visually depicts the number of studies included and excluded at each stage along with specific reasons for exclusion [72].

B. RESEARCH QUESTIONS

In this study, six key research questions have been defined, aimed at exploring various dimensions of drug-drug interactions (DDIs) and evaluating their potential in practical applications. These questions cover a wide range of topics, from the performance optimization of DDI prediction models to privacy and security advantages, data scalability, and applicability in resource-constrained environments. Each research question focuses on a different aspect of DDIs, providing a comprehensive framework for understanding these interactions and exploring their future potential. The research questions outlined below form the cornerstone of our study, addressing critical issues related to the development, implementation, and practical outcomes of DDIs, thereby guiding the direction of our research.

TABLE 3. Deep learning-based methods for drug–target interaction prediction.

Reference	Year	Model	Type	Metrics
Öztürk et al. [50]	2018	CNN, CNN	Classic	MSE, CI
Zheng et al. [51]	2018	CNN-LSTM	Classic	AUC, Pre
Lee et al. [52]	2019	CNN, CNN	Classic	Sen, Spe, Pre, Acc, F1
Jiang et al. [53]	2020	GNN, GNN	Classic	MSE, CI, Pearson
Nguyen et al. [54]	2021	CNN, GNN	Classic	MSE, CI
Yang et al. [55]	2022	CNN, CNN	Classic	MSE, CI
Mukherjee et al. [56]	2022	LSTM, GCN	Classic	MSE, CI
Zhang et al. [57]	2022	GCN, GCN	Classic	MSE, CI, Pearson
Wu et al. [58]	2022	CNN, GCN	Classic	AUROC, F1
Peng et al. [59]	2020	CNN	Interactive	AUROC, AUPRC, ACC
Wang et al. [60]	2022	CNN	Interactive	ACC, F1, AUROC
Li et al. [61]	2022	GCN	Interactive	ACC, Recall, Spe, AUC
Jimenez et al. [62]	2018	3D CNN	3D space	RMSD, R
Li et al. [63]	2021	SIGN	3D space	RMSE, MAE, SD, R
Zhou et al. [64]	2023	Uni-Mol	3D space	RMSD
Meng et al. [65]	2024	Graph Transformer + CNN	Classic / Transformer	AUROC, AUPR
Zeng et al. [66]	2024	Transformer (cross-attention)	Transformer	AUROC, F1
Liu et al. [67]	2024	GNN + Transformer	Interactive / Transformer	Precision, Recall
Zhang et al. [68]	2024	GAT + attention	Classic / Heterogeneous	AUROC, AUPRC
Liu et al. [69]	2024	Hierarchical Graph Representation	Classic / Graph	ROC, F1
Meng et al. [70]	2024	Token-level Fusion + PLM	Transformer / PLM	Benchmark Acc
Author et al. [71]	2025	GM-Transformer + LLM	Transformer / LLM	MCC, AUROC

CNN = convolutional neural network, GCN = graph convolutional network, GNN = graph neural network, LSTM = long short-term memory, RMSE = root mean squared error, MAE = mean absolute error, SD = standard deviation, R = correlation coefficient, Sen = sensitivity, Spe = specificity, ACC = accuracy, AUC = area under the ROC curve, AUPR = area under the precision–recall curve, AUPRC = area under the precision–recall curve, LLM = large language model.

RQ1: How do different datasets and sources (e.g., biological data, clinical data) impact the accuracy of AI-based drug interaction prediction models?

RQ2: Which model architectures are most effective for different types of data (molecular, clinical, literature)?

RQ3: What are the most suitable methods for modeling drug interactions using AI techniques?

RQ4: What are the most commonly used AI models in drug discovery and drug-drug interactions?

RQ5: How can the success rates of deep learning-based approaches be compared to traditional machine learning approaches?

RQ6: What AI techniques are used to address data imbalance and identify rare interactions in the case of insufficient or imbalanced data?

C. SEARCH STRATEGY

The method followed to conduct a systematic literature review in this study is explained in detail. The databases used for the review process were PubMed, Scopus, Web of Science, and IEEE Xplore. The search covers publications from the years 2010 to 2025. The final database search was conducted on December 20, 2025. Search queries were systematically constructed using the Boolean Query method. Only documents written in English were included in the study. The types of documents reviewed were research articles, conference papers, and review articles. This structured approach aims to ensure that the literature review is comprehensive, reproducible, and conducted in accordance with academic standards.

1) KEYWORDS

This section describes the search strategy and keywords used in our systematic literature review on drug-drug interaction prediction. To ensure methodological transparency, each keyword was systematically derived by decomposing the Research Questions (RQ1–RQ6) into their core thematic components, such as interaction types, AI architectures, and evaluation benchmarks. This mapping process, as illustrated in Table 4, ensures that the search queries address the study's objectives both broadly and with a focused approach. The process is divided into primary and secondary keywords, where primary terms define the research domain and secondary terms capture the methodological and evaluative dimensions. These keywords were selected in alignment with our research questions and aim to ensure a comprehensive search of the relevant literature.

a: PRIMARY KEYWORDS

Drug-drug interaction*: Focuses on drug-drug interactions, which are the central topic of our research. The use of the asterisk (*) allows for the inclusion of derivatives such as "interaction" and "interactions."

b: SECONDARY KEYWORDS

Adverse drug events: Covers studies related to adverse effects associated with DDIs and focuses on safety issues.

Machine Learning: Targets studies related to predictive models used for DDI analysis and is one of the key areas of our systematic review.

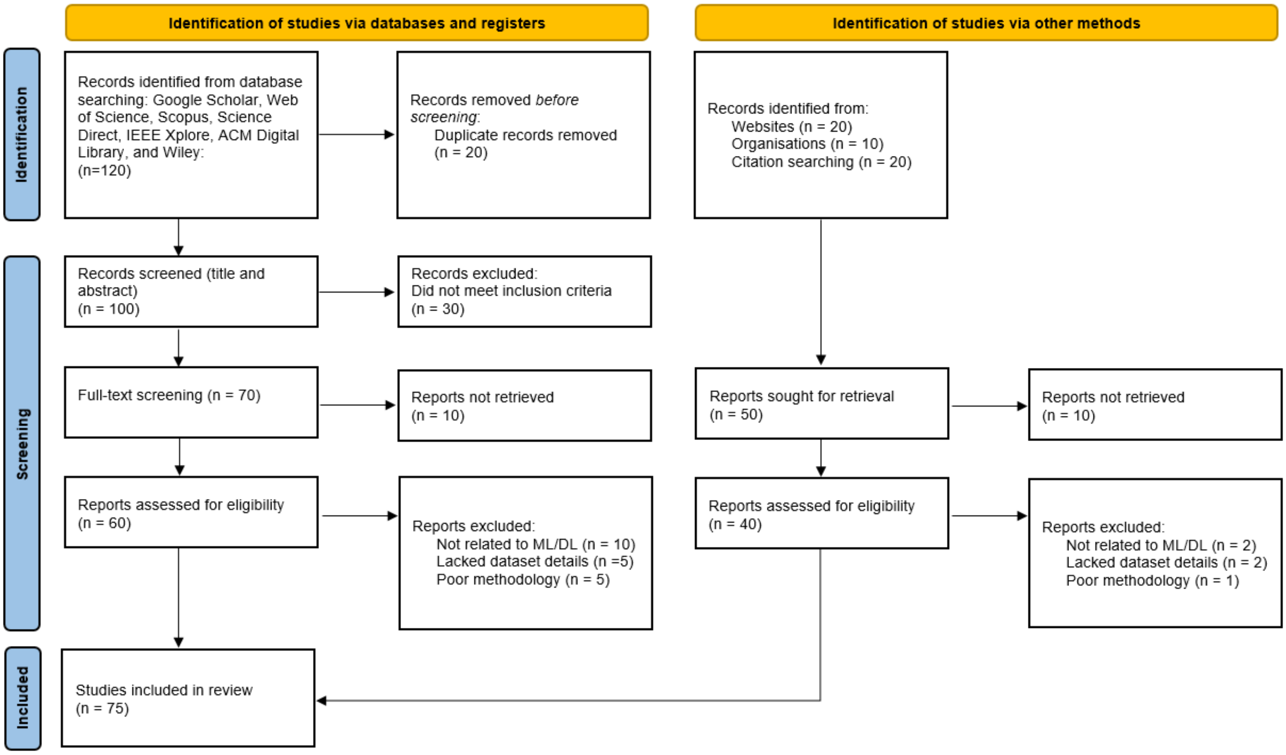

FIGURE 4. PRISMA 2020 Flow Diagram. Flow of the number of records identified, screened, assessed for eligibility, and included in the systematic review.

Deep Learning: Includes studies utilizing advanced neural network techniques and sheds light on methodological innovations.

Computational pharmacology: Addresses drug interactions within the context of computational biology, covering in silico methods used for DDI prediction.

Artificial Intelligence in drug discovery: Focuses on the applications of Artificial Intelligence in the drug development process and examines DDI prediction in a broader context.

Drug similarity models: Includes studies utilizing similarity-based methods, which are one of the fundamental approaches in DDI research.

Privacy-focused Artificial Intelligence: Includes studies addressing ethical and privacy concerns in the use of medical data.

These keywords were combined to create various search strings using Boolean operators (AND, OR). Our strategy aims to identify the most current and relevant studies that illuminate the challenges and methods related to DDI prediction. The integration of primary and secondary keywords has allowed us to approach the research area both from a broad perspective and with a focused approach.

2) QUERY STRING

A comprehensive search strategy for the drug interaction literature has been carefully designed to cover the entire body of literature while maintaining the precision of the results. Custom search strings were developed to meet the unique syntax requirements and search capabilities of each database, while the core concepts of our research were preserved. In this context, the focus has been placed on results that include terms such as "drug-drug interactions" or "drug interaction prediction" along with methods like "deep learning," "machine learning," or "AI," and performance metrics such as "model performance," "AUC," or "accuracy" (Table 5).

3) INCLUSION CRITERIA

I1: Studies published between 2010 and 2025

This criterion is established to ensure a comprehensive evaluation of the technological trajectory within the field. Literature published within this period reflects foundational developments, innovations, and trends, providing a robust representation of advancements in technology and the research field from early baselines to contemporary state-of-the-art models.

I2: Articles focusing on AI and deep learning-based drug interaction models

This criterion includes only studies examining drug interaction models that utilize Artificial Intelligence (AI) and Deep Learning techniques. AI and Deep Learning provide groundbreaking methods for predicting and analyzing drug interactions in contemporary research, making studies in this area the focal point.

I3: Studies reporting performance metrics

Performance metrics are essential for evaluating the effectiveness and accuracy of models. Therefore, only studies that explicitly report the success or performance of the models will be considered.

TABLE 4. Systematic mapping of research questions to thematic search keywords and strategy components.

Research Question (RQ)	Core Thematic Component	Derived Keywords
RQ1 & RQ2 (Impact of Datasets and Architectures)	Data sources and model frameworks	Drug similarity models, Machine Learning, Deep Learning, Computational pharmacology
RQ3 & RQ4 (Methods and AI Models)	AI-driven DDI prediction paradigms	Drug-drug interaction*, Artificial Intelligence in drug discovery
RQ5 (Success Rate Comparison)	Performance evaluation/benchmarking	Machine Learning, Deep Learning, Adverse drug events
RQ6 (Data Imbalance and Rare Interactions)	Data scarcity and robustness	Machine Learning, Deep Learning, Privacy-focused Artificial Intelligence

DDI = drug–drug interaction, RQ = research question, AI = artificial intelligence. The asterisk (*) is used as a wildcard to include derivatives such as "interaction" and "interactions".

TABLE 5. Search queries used in different databases for drug interaction prediction.

Database	Search Query
IEEE Xplore	("drug-drug interactions" OR "drug interaction prediction") AND ("deep learning" OR "machine learning" OR "AI") AND ("model performance" OR "AUC" OR "accuracy")
Scopus	(TITLE-ABS-KEY("drug-drug interactions" OR "drug interaction prediction")) AND (TITLE-ABS-KEY("deep learning" OR "machine learning" OR "AI")) AND (TITLE-ABS-KEY("model performance" OR "AUC" OR "accuracy"))
PubMed	("drug-drug interactions"[MeSH Terms] OR "drug interaction prediction") AND ("deep learning" OR "machine learning" OR "AI") AND ("model performance" OR "AUC" OR "accuracy")
arXiv	all:"drug-drug interactions" OR all:"drug interaction prediction" AND all:"deep learning" OR all:"machine learning" OR all:"AI" AND all:"model performance" OR all:"AUC" OR all:"accuracy"
Google Scholar	"drug-drug interactions" OR "drug interaction prediction" AND "deep learning" OR "machine learning" OR "AI" AND "model performance" OR "AUC" OR "accuracy"

AI = artificial intelligence, DDI = drug–drug interaction, ML = machine learning, DL = deep learning, AUC = area under the curve, PubMed = public medical database, IEEE = Institute of Electrical and Electronics Engineers.

I4: Articles published in peer-reviewed journals

Articles published in peer-reviewed journals guarantee scientific validity and quality. These publications provide more reliable and accurate information, and as such, focus will be placed on such works.

I5: Articles with accessible full text

Studies with accessible full texts allow for detailed analysis, access to datasets, and methodologies. This enables a better understanding of the accuracy and reliability of the examined models and data.

4) EXCLUSION CRITERIA

E1: Studies using methods other than AI

This criterion excludes studies based on older or traditional methods that do not utilize modern and advanced techniques such as AI and deep learning. The absence of such techniques in a study indicates that it does not fall within the targeted scope of this review.

E2: Non-clinical or non-experimental prediction models

Prediction models that are not based on clinical and experimental studies are theoretical or simulation-based rather than grounded in real-world applications. Therefore, only studies supported by clinical and experimental data will be considered.

E3: Non-literature sources

Non-literature sources, such as blog posts or white papers, are generally lacking scientific validation and reliability. These sources are considered less trustworthy compared to scientific research and will be excluded.

E4: Publications in languages other than English

Studies published in languages other than English can complicate literature review and data analysis. To ensure broader accessibility, English-language publications have been prioritized.

E5: Studies that do not provide details of the dataset or model

Studies that do not present detailed information about the model and dataset are limited in terms of reproducibility and reliability. Therefore, studies that do not provide such details will be excluded.

5) QUALITY SCORE

The quality of the articles can be assessed based on the following criteria:

QS1: Model Performance: Are the performance metrics used clearly reported? (0–2 points)

QS2: Dataset: Is the dataset clearly defined and reproducible? (0–2 points)

QS3: Method Transparency: Is the model and methodology described in detail? (0–2 points)

QS4: Reliability of Results: Are the results statistically validated? (0–2 points)

QS5: Relevance: Is the study aligned with the main review question? (0–2 points)

In this study, articles scoring between 8 and 10 out of a maximum of 10 points were classified as high-quality studies. This threshold was chosen because a score of 8 or above indicates that a study met at least four out of five key quality criteria at a high level.

IV. RESULTS

In this section, we summarize the key findings obtained from the systematic review of 75 studies focusing on AI-based drug–drug interaction (DDI) prediction. The results are organized according to the predefined research questions (RQ1–RQ6), covering data sources, feature representation strategies, model architectures, evaluation metrics, and approaches for addressing data imbalance and rare interactions. Figures and tables provide a comparative overview of methodological trends and performance outcomes across studies.

A. MAIN FINDINGS OF INCLUDED STUDIES

In this section, the general observations derived from the systematic review and the specific findings related to each research question are presented. The temporal distribution of the literature is illustrated in Fig. 5, which demonstrates a consistent upward trend in the volume of publications from 2010 through 2025. This steady growth highlights the intensifying academic interest and the rapid evolution of computational methodologies—transitioning from traditional machine learning to advanced deep learning paradigms—in the field. Notably, as the final database search was conducted on December 20, 2025, the data provide a comprehensive representation of the annual research output for 2025, ensuring that the latest state-of-the-art developments are fully incorporated into the analysis. Furthermore, a geographical analysis of the contributing authors' affiliations is provided in Fig. 6. As depicted, the United States and China emerge as the dominant contributors to the global research landscape, significantly outpacing other regions in terms of document volume. Following these bibliometric insights, the subsequent subsections provide a detailed thematic analysis of the observations concerning each predefined research question.

Below are the revised summaries of the six research questions (RQ-1 to RQ-6), integrating both descriptive

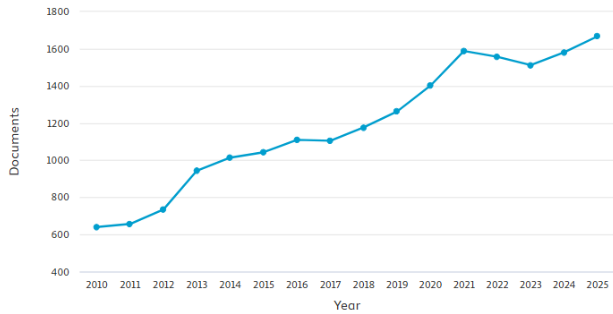

FIGURE 5. Temporal distribution of the selected primary studies from 2010 to 2025, highlighting the consistent growth in DDI prediction research.

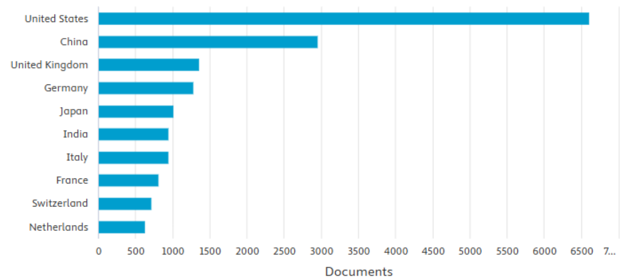

FIGURE 6. Geographical distribution of research contributions by country, identifying the global leaders in AI-driven pharmacological studies.

findings and critical evaluation of methodological rigor, biases, and limitations.

RQ-1: How do different datasets and sources (e.g., biological data, clinical data) impact the accuracy of AI-based drug interaction prediction models?

Different datasets and sources influence the accuracy of AI-based drug interaction prediction models in various ways. These effects can be summarized based on the quality, scope, and source of the data used:

Clinical Data: In the literature, FDA's (U.S. Food and Drug Administration) AERS (Adverse Event Reporting System) / FAERS (FDA Adverse Event Reporting System) reports and structured databases (e.g., DrugBank) are commonly used. These clinical data, due to the large sample sizes derived from raw reports, enhance the generalizability of models, and performance improvements can be achieved when proper preprocessing and data standardization techniques (e.g., Modified Skip-Gram method) are applied [73], [74].

Biological Data: Biological data sources, such as porcine tissue explants, play a significant role in improving model accuracy, as they reflect real physiological conditions. These types of data allow more realistic modeling of drug-transporter interactions, especially in cases where cell or animal models fall short [75].

Statistical and Other Approaches: In clinical data analysis, statistical methods such as propensity score-matched three-component mixture models are used to reduce confounding effects, thereby minimizing the likelihood of false-positive results and enhancing the reliability of the model. Additionally, some studies have optimized model accuracy by integrating various data sources, combining information from both clinical and biological datasets.

Critical Appraisal: A critical appraisal of current methodologies reveals significant limitations regarding intrinsic dataset bias. Most widely used datasets are predominantly comprised of pharmaceutical data from high-income countries, which creates a substantial overrepresentation that constrains the global applicability and generalizability of AI-driven predictive models. This geographic concentration suggests that models may not perform with equal reliability across diverse global populations or for drugs primarily used in low-to-middle-income regions. Furthermore, the quality of data sources varies considerably; reliance on self-reported data introduces reporting bias, while the absence of standardized preprocessing protocols complicates cross-study comparisons and potentially inflates performance metrics. Additionally, the field may be influenced by publication bias, as studies reporting substantial accuracy gains through dataset integration are more likely to be prioritized for dissemination.

Beyond geographic bias, the precision of AI models is closely tied to the quality of the underlying data. Significant disparities exist in the quality of data sources utilized, and the lack of standardized preprocessing techniques further complicates direct comparisons between studies. Such variability in data management and experimental protocols often results in the inflation of performance metrics, which may undermine the reliability of the findings. Consequently, while high accuracy rates are frequently reported, they must be interpreted with caution due to these methodological inconsistencies and the potential for over-optimistic results.

Table 6 summarizes the primary databases utilized in drug-drug interaction (DDI) prediction. DrugBank serves as a cornerstone, providing exhaustive data on drug characteristics and their corresponding target proteins. Complementing this, SIDER and TWOSIDES offer extensive datasets focused on adverse drug effects, while PubChem acts as a vital repository for chemical structures and bioactivity results. To facilitate a systemic understanding, KEGG offers insights into metabolic pathways and enzyme-protein relationships. Furthermore, Medline supports literature-based discovery through its vast collection of biomedical abstracts. Collectively, these resources constitute the foundational infrastructure for machine learning and bioinformatics-driven drug discovery.

RQ-2: Which model architectures are most effective for different types of data (molecular, clinical, literature)?

The findings obtained from the reviewed papers for determining the most effective model architectures for different data types (molecular, clinical, literature) are as follows:

1) Effective Model Architectures for Molecular Data

Molecular data are primarily used for predicting drug-target interactions. The following models stand out in this field:

MolTrans (Molecular Interaction Transformer): This transformer-based model makes predictions by considering the sub-structural properties of drugs and proteins. MolTrans provides higher accuracy and explainability by utilizing large-scale unlabeled biomedical data [10].

Graph Convolutional Networks (GCN) and Deep Neural Networks (DNN): The GCN-DTI model predicts drug-target interactions by constructing a network where drugs and proteins are represented as nodes. This model is particularly successful in identifying both known drug-target interactions and potential new interactions to be discovered [76].

2) Effective Model Architectures for Clinical Data

Clinical data include patient records, genetic information, and drug interactions. The following approaches have been effective for this type of data:

Multi-Feature Deep Learning Models: The SSF-DDI (Sequence-Substructure Fusion for Drug-Drug Interaction) model predicts clinical drug interactions by using both sequence-based and substructure-based features of drugs. It enhances prediction accuracy by focusing on key substructural features through the SGFE (Subgraph-based Graph Feature Extraction) module [77].

Cross-Attention Mechanism Networks (MCANet): This model uses a cross-attention mechanism to consider different aspects of drugs and proteins while analyzing drug-target interactions. As a result, it achieves higher accuracy in predicting clinical drug interactions.

3) Effective Model Architectures for Literature Data

Text mining and Natural Language Processing (NLP) techniques are used to extract information from biomedical literature. Machine Learning methods, such as Support Vector Machines (SVM) and Random Forests, are employed to identify drug-target relationships in biomedical texts. However, the effectiveness of these methods depends on the quality and diversity of the data in the literature.

Critical Appraisal: These architectures are often optimized for specific data types, which raises concerns about their ability to generalize to other modalities. The complexity of certain deep learning models increases the risk of overfitting when data are scarce, and reliance on benchmark datasets may overstate real-world performance. Furthermore, few studies evaluate the same architecture across multiple heterogeneous datasets, limiting conclusions about overall robustness.

RQ-3: What are the most suitable methods for modeling drug interactions using AI techniques?

Artificial Intelligence methods used to model drug interactions are generally classified into three main categories:

1) Deep Learning-Based Methods

Deep learning is a commonly used approach for predicting drug interactions. In these methods, the molecular structures of drugs and biomedical knowledge networks are combined

TABLE 6. Overview of common databases and their entity characteristics for drug interaction prediction.

Database	Entities	Brief Description
DrugBank	Drugs, Targets, Proteins	Contains detailed information about drugs and their associated targets or proteins.
SIDER	Drugs, ADRs	Provides data on adverse drug reactions for numerous marketed drugs.
TWOSIDES	Drugs, ADRs	Includes information about adverse drug reactions and drug pair associations.
PubChem	Structure	Open repository containing chemical structures and results from biological assays.
KEGG	DDIs, Proteins	Describes metabolic pathways linking metabolites with proteins and enzymes.
Medline	Abstract	Database of abstracts from biomedical and life sciences journal articles.

DDI = drug–drug interaction, KEGG = Kyoto Encyclopedia of Genes and Genomes, MEDLINE = Medical Literature Analysis and Retrieval System, ADR = adverse drug reaction.

to make more accurate predictions. Models that perform data fusion at different scales, in particular, address the challenges of limited labeled data and demonstrate high performance in both binary classification and multi-label prediction tasks [78].

Additionally, Long Short-Term Memory (LSTM)-based deep learning models stand out as an effective method for modeling drug-target interactions. LSTM models combine the evolutionary information of proteins with the molecular structures of drugs to make high-accuracy predictions. For example, in one study, an LSTM-based model was developed by combining the molecular substructure properties of drugs and protein sequences, enabling the learning of long-term dependencies, and achieving high performance [79].

2) Graph-Based Methods

Graph-based approaches offer powerful tools for understanding the relationships between drugs and biological entities. These methods use networks containing various biomedical data, such as drugs, proteins, diseases, and side effects, to predict drug interactions [80].

Graph Convolutional Networks (GCN) and Knowledge Graph Embedding (KGE) methods provide successful results in identifying drug-drug and drug-target interactions. These models improve prediction accuracy by utilizing the multifaceted relationships derived from biomedical knowledge graphs. Particularly, graph-based models trained on heterogeneous biomedical networks provide significant advantages in modeling drug-protein and drug-disease relationships.

3) Machine Learning and Feature Engineering-Based Methods

Traditional machine learning techniques are also used for predicting drug interactions. Methods such as Support Vector Machines, Random Forests, and Collective Matrix Factorization (CMF) analyze features extracted from biomedical data to determine drug-target interactions [81].

Figure 7 illustrates how matrix factorization methods are used to predict drug-drug interactions (DDIs) by uncovering latent relationships between pharmaceutical entities. Initially, a matrix D $(n \times n)$ is constructed, which contains known DDIs, where 1s represent known interactions and 0s represent unknown interactions. The matrix D is then decomposed into İwo matrices, A $(n \times \ell)$ and B $(\ell \times n)$, using various matrix factorization techniques. This decomposition process aims to map drugs into a lower-dimensional latent space, capturing hidden biological and chemical patterns that are not immediately apparent in the raw data. When these two matrices are multiplied, the matrix D_{pre} $(n \times n)$ is obtained, which predicts potential drug interactions by estimating values for the previously unknown (0) entries. This method is widely used in machine learning and recommendation systems to handle data sparsity and identify novel interactions based on shared latent features.

These methods can be particularly effective when working with small datasets, but their ability to model complex relationships is limited compared to deep learning and graph-based approaches. However, some studies have shown that the performance of drug-target prediction models can be improved by using feature engineering and dimensionality reduction techniques.

The most suitable AI method depends on the type of data used and the objectives of the research. When working with large datasets, deep learning models provide higher accuracy. For making predictions using information derived from biomedical networks, graph-based methods are preferred. In applications where limited data is available or interpretable models are required, traditional machine learning methods may be effective. Overall, the best results for predicting drug interactions are achieved using methods that effectively integrate biomedical information and can process diverse data sources.

Table 7 summarizes machine learning-based drug-drug interaction prediction methods under different categories. Among traditional methods, classical similarity-based approaches are based on the assumption that similar drugs may have similar interactions, while traditional classification approaches categorize drug pairs as interacting or non-interacting. Network diffusion-based methods treat drugs as nodes and interactions as edges, attempting to predict unknown connections. Within this category, link prediction methods extract unknown connections from existing links in a network, while graph embedding methods convert the network into a lower-dimensional space to learn node representations. Matrix factorization decomposes the known drug-drug interaction matrix into smaller components under specific constraints and reconstructs them to predict new interactions. The ensemble-based approach combines

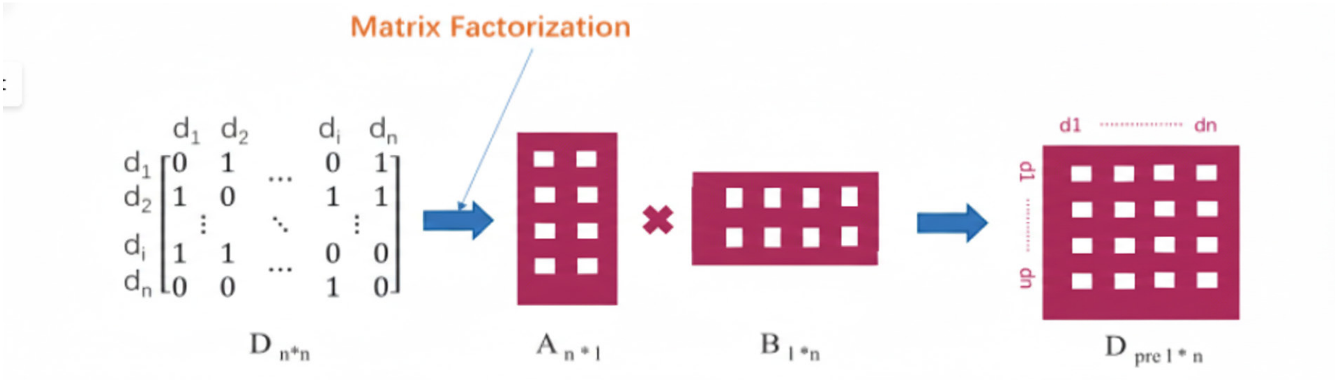

FIGURE 7. Transformation of the sparse DDI adjacency matrix into a predictive interaction matrix via latent space decomposition.

TABLE 7. Systematic classification and operational mechanisms of machine learning-based paradigms for drug–drug interaction prediction.

Category	Method	Description
Traditional similarity	[82]-[84]	Drug A and B interact to produce an effect, likely replicated by similar drugs.
Traditional classification	[85]-[87]	The prediction task is a binary classification using drug interaction and non-interaction pairs.
Network diffusion-Link prediction	[88]-[92]	Drugs are modeled as nodes, with interactions represented as edges, to predict unknown interactions using label propagation, RLS and graph traversal.
Network diffusion-Graph embedding	[93]-[99]	The graph is mapped to a low-dimensional space preserving structure, enabling automatic learning of node representations for prediction.
Matrix factorization	[100]-[105]	Decomposes the known DDI matrix into latent matrices based on similarity, then reconstructs them to generate new interaction predictions.
Ensemble-based approach	[106]	Combines multiple methods to predict unknown DDIs.
Based on literature	[107]-[109]	Text mining and statistical methods extract relationships from unstructured data, which are then used by machine learning models to predict new DDIs.

DDI = drug–drug interaction, RLS = regularized least squares.

multiple methods to make stronger predictions. Finally, literature-based approaches extract drug relationships from unstructured data sources using statistical or text mining techniques and predict unknown drug interactions using machine learning. Each of these methods has distinct advantages, demonstrating how various machine learning approaches are applied in drug interaction prediction.

Critical Appraisal: LSTM-based models and multi-scale feature fusion techniques effectively process heterogeneous biomedical data, while graph-based methods such as GCN and Knowledge Graph Embedding excel at leveraging relational structures. Traditional ML methods, including SVM and Random Forests, remain relevant in scenarios with small datasets or when interpretability is required. However, there is often a trade-off between performance and interpretability, with the most accurate models frequently being 'uninterpretable models' that hinder clinical trust and pose practical barriers to their adoption in real-world healthcare settings. Evaluation methodologies vary widely, with inconsistent use of performance metrics like AUROC, MSE, and Accuracy. This lack of standardization complicates direct comparison across studies. Moreover, negative or inconclusive results are rarely reported, which can skew the perceived superiority of certain methods.

Q-4:What are the most commonly used AI models in drug discovery and drug-drug interactions?

To identify the most commonly used artificial intelligence models in drug discovery and drug-drug interaction prediction, a review of the literature was conducted. The findings are summarized below:

1. **Deep Neural Networks (DNN)**

DNNs are widely used for drug-drug interaction prediction. For example, the DeepDDI model utilizes deep learning techniques to predict interactions by analyzing the chemical structures of drugs. The Dependency-based Convolutional Neural Network (DCNN) has been used to predict drug-drug interactions (DDIs) by analyzing text. This text mining-based method was developed to extract relationships from biomedical literature [110].

2. **Knowledge Graph Neural Networks (KGNN)**

KGNN is a model developed for drug interaction prediction using relationships derived from knowledge graphs. This method can determine interactions with high accuracy by extracting structural relationships of drugs [111].

3. **Convolutional Neural Networks (CNN)**

DTI-CNN is a CNN model used to predict drug-target interactions (DTIs). This model predicts interactions using features extracted from heterogeneous biological networks [59].

4. **Feature Representation Learning ve Autoencoders**

Denoising Autoencoder (DAE) is a method used to reduce the high dimensionality of drug and protein interaction data and to identify essential features. It is frequently used in CNN-based drug-target interaction prediction models.

5. **Similarity-Based Methods and Machine Learning**

Similarity Network Fusion (SNF) is another widely used approach for predicting drug-drug interactions. This method enhances prediction accuracy by integrating the chemical, target-based, side-effect, and biological similarities of drugs.

Traditional machine learning methods such as Logistic Regression, Support Vector Machines, and Decision Trees have also been applied for drug interaction prediction.

In conclusion, among the most commonly used artificial intelligence models in drug discovery and drug-drug interactions are Deep Neural Networks (DNN, CNN, KGNN), Knowledge Graphs (KGNN), Autoencoders (DAE), and Similarity-Based Models (SNF, traditional machine learning methods).

Critical Appraisal: Deep learning models often dominate the literature due to their popularity, creating a form of "hype bias" that may overshadow simpler yet effective alternatives. Direct, head-to-head comparisons between model types on identical datasets are uncommon, making it difficult to objectively determine superiority. Furthermore, DL performance is strongly dependent on the availability of large, high-quality labeled datasets, which are not always accessible in real-world settings.

RQ-5: How can the success rates of deep learning-based approaches be compared to traditional machine learning approaches?

The findings obtained from three studies comparing the success rates of deep learning-based approaches with traditional machine learning approaches are as follows:

1) Deep Learning vs. Traditional Machine Learning

Traditional machine learning methods are typically based on feature engineering and utilize manually extracted molecular features. In contrast, deep learning models can directly create complex representations from raw data. While they often report higher accuracy rates, these findings must be interpreted with caution. The perceived superiority of DL is not universal; rather, it represents a trend observed primarily in large-scale datasets. In contrast, traditional ML models often demonstrate competitive or even superior performance in data-scarce scenarios. Furthermore, due to the lack of direct, head-to-head comparisons utilizing the same datasets and standardized protocols, any definitive claim regarding the inherent superiority of one model type over another remains logically inconclusive.

Deep Learning models, such as Graph Neural Networks (GNN), have made significant progress in fields like drug discovery, enabling better modeling of molecular relationships. However, a key disadvantage of these models is the need for large labeled datasets and challenges in generalization [112].

2) Comparison of Performance

Deep learning models developed using Convolutional Neural Networks (CNN) have achieved higher accuracy in predicting drug-drug interactions compared to traditional methods [113].

Graph Neural Networks (GNN)-based models have demonstrated higher AUC-ROC and lower error rates in predicting drug-target interactions compared to traditional machine learning methods.

The MOVE framework (Multi-View Optimal Variational Embedding) has successfully uncovered drug-omics relationships more effectively than traditional statistical methods [114].

3) Advantages of Deep Learning

Deep learning models are more effective in handling large and complex datasets. These models possess the ability to better integrate and analyze multi-dimensional and heterogeneous data. This capability enables the processing of information from diverse data sources, resulting in more comprehensive and accurate outcomes.

Compared to traditional machine learning methods, deep learning models are less reliant on feature engineering. Traditional methods typically depend on manually conducted analysis to identify the correct features, whereas deep learning models can extract meaningful representations from raw data. This feature allows deep learning to perform information extraction in a more automated and efficient manner.

4) Challenges of Deep Learning

Deep learning models often require large amounts of labeled data. However, in areas such as drug discovery, the collection of such data can be time-consuming and costly. The lack of labeled data can negatively impact the performance of deep learning models, making it more difficult for the model to produce accurate results. Furthermore, deep learning models trained on small datasets are at risk of overfitting. This occurs when the model adapts too closely to the training data, limiting its ability to generalize to new, larger datasets.

In conclusion, deep learning approaches generally offer higher success rates than traditional machine learning in fields involving large datasets and complex relationships. However, due to challenges like the need for large datasets and generalization issues, hybrid approaches are commonly preferred.

Critical Appraisal: DL models, including CNNs and GNNs, generally outperform traditional ML in AUROC, accuracy, and regression metrics, especially for large, heterogeneous datasets. In smaller datasets, ensemble ML models can perform competitively or even better. Despite these trends, several methodological issues temper claims of DL superiority. First, the field is likely influenced by publication bias, as studies reporting positive outcomes for deep learning are more frequently disseminated, potentially distorting the perceived efficacy of these approaches. Moreover, the current literature exhibits a 'hype bias,' where the prominence of complex deep learning models may eclipse the viability of simpler, yet highly effective, traditional machine learning alternatives that may perform equally well in data-scarce scenarios. Deep learning models necessitate expansive labeled datasets, which are frequently limited in the context of drug discovery. This scarcity elevates the risk of overfitting, wherein a model demonstrates strong performance on training data yet fails to generalize effectively to novel, unseen drug combinations. Furthermore, many studies predominantly utilize internal cross-validation without incorporating independent test sets, a practice that can artificially enhance performance metrics and obscure these generalization issues. Moreover, common metrics like AUROC and Accuracy can mask poor performance on minority classes, making it essential to report precision, recall, and F1-score. Finally, inconsistent validation strategies (e.g., random split vs. leave-drug-out) make cross-study comparisons less reliable. The lack of a standardized evaluation framework—where disparate validation protocols and varying metrics like AUROC, accuracy, and F1-score are applied inconsistently—impedes the ability to definitively ascertain which architectural framework is inherently superior.

RQ-6: What AI techniques are used to address data imbalance and identify rare interactions in the case of insufficient or imbalanced data?

Various artificial intelligence techniques are used to address data imbalance and identify rare interactions in cases of insufficient or imbalanced data:

1. Techniques Used to Address Data Imbalance

Data augmentation is a method aimed at improving model performance by leveraging missing or unreliable experimental data. In some studies, approaches are proposed to identify reliable biological replicates from less reliable experiments and include them in the training dataset. This method enhances the model's robustness and allows for more accurate predictions by increasing the number of reliable data points [115].

Several methods are used to solve the missing data problem. Techniques such as K-Nearest Neighbors (kNN), Singular Value Decomposition (SVD), Principal Component Analysis (PCA), and Support Vector Regression (SVR) are effective in predicting missing data and completing the dataset. These methods ensure that missing data are accurately predicted, improving the accuracy of analyses and preventing errors caused by missing data.

2. Techniques Used to Identify Rare Interactions

Graph Convolutional Networks (GCN) are an effective method for identifying rare drug-drug interactions by learning the topological relationships between drugs. GCN models facilitate the prediction of low-frequency interactions by analyzing these relationships. As a result, interactions that were previously difficult to discover can be predicted more accurately by considering the structural connections between drugs [116].

Deep fusion strategies improve the identification of rare interactions by enabling deep learning models to combine both the properties of drugs and the topological relationships between them. Specifically, using multi-layered data fusion to learn features at different levels enhances the model's ability to capture rare interactions. This approach enables more comprehensive analysis of drug interactions and leads to more accurate predictions.

Various statistical and machine learning methods are also used for identifying drug interactions. These methods include Semi-Negative Matrix Factorization (semi-NMF), interaction profile fingerprint analysis, and Bayesian networks. Semi-Negative Matrix Factorization is effective for completing missing data and discovering hidden interactions, while interaction profile fingerprint analysis helps identify the interaction properties of drugs. Bayesian networks offer a more dynamic modeling approach using a probability-based framework to uncover drug interactions. These methods enable more accurate and efficient modeling of drug interactions [117].

In conclusion, data augmentation and missing data prediction techniques are commonly used to address data imbalance. For identifying rare interactions, GCN, deep learning-based fusion methods, and statistical analysis techniques provide effective solutions.

A comparative analysis of machine learning and deep learning approaches for drug–drug interaction (DDI) prediction reveals a significant evolutionary shift from structure-based and knowledge graph-based (KG-based) models toward Transformer architectures and Large Language Models (LLMs). As synthesized in Table 8, earlier methods primarily relied on structural similarities or graph-based embeddings (e.g., DeepDDI, SumGNN), whereas the most recent advancements from 2024 and 2025 demonstrate a transition toward attention-based mechanisms such as MolFormer and cross-modal strategies.

Performance metrics presented in Table 9 confirm that these architectural transitions have led to substantial improvements in predictive accuracy. While 2021-era models typically reported AUROC values in the 0.91–0.99 range, the TxGemma model has established a new state-of-the-art benchmark with an AUROC of 0.981 and a Macro-F1 score of 0.962. Furthermore, the simultaneous high performance in Accuracy (0.952) and AUPR (0.935) across these new-generation models indicates a superior ability to handle class imbalance within multi-class classification tasks compared to traditional ensemble-based ML models.

TABLE 8. Comparative performance metrics of advanced computational models for multi-task drug interaction prediction.

Study	Year	Method/Algorithm	Validation Scheme
DeepDDI [112]	2018	Structure-based (Multi-class)	training, validation and test sets
DDIMDL [113]	2019	Structure-based (Multi-class)	5-fold cross-validation
Hou et al. [114]	2019	Structure-based (Multi-class)	training, validation and test sets (6:2:2)
SSI-DDI [115]	2021	Structure-based (Multi-class)	training, validation and test sets
MDNN [116]	2021	KG-based (Multi-class)	5-fold cross-validation
MUFFIN [117]	2021	KG-based (Multi-class/label)	5/6-fold cross-validation
SumGNN [118]	2021	KG-based (Multi-class/label)	training, validation and test sets (7:1:2)
META-DDIE [119]	2022	Structure-based (Multi-class)	5-fold cross-validation
MDDI-SCL [120]	2022	Structure-based (Multi-class)	5-fold cross-validation
TBPM-DDIE [121]	2022	Structure-based (Multi-class)	6-fold cross-validation
DM-DDI [122]	2022	Network-based (Multi-class)	5-fold cross-validation
MDF-SA-DDI [123]	2022	Network-based (Multi-class)	6-fold cross-validation
STNN-DDI [124]	2022	Structure-based (Multi-class/label)	5/10-fold cross-validation
GMPNN-CS [125]	2022	Structure-based (Multi-class/label)	3-fold cross-validation
LaGAT [126]	2022	KG-based (Multi-class)	4-fold cross-validation
DeepMDDI [127]	2022	Network-based (Multi-class)	5-fold cross-validation
R2-DDI [128]	2023	Structure-based (Multi-class/label)	training, validation and test sets (3:1:1)
DSN-DDI [129]	2023	Network-based (Multi-class/label)	3/4-fold cross-validation
ACDGNN [130]	2023	Network-based (Multi-class)	training, validation and test sets (6:2:2)
MCFF-MTDDI [131]	2023	KG-based (Multi-class/label)	4/5-fold cross-validation
MolFormer [132]	2024	Transformer-based (Linear Attention)	5-fold cross-validation
DrugGPT [133]	2024	GPT-based Autoregressive model	training, validation and test sets
TxGemma [134]	2025	Agentic LLM for therapeutics	independent external test sets
DDI-Transformer [135]	2025	Cross-Modal Attention	10-fold cross-validation

DDI = drug-drug interaction, KG = knowledge graph, CV = cross-validation, DL = deep learning, GNN = graph neural network, LLM = large language model, GPT = generative pre-trained transformer, Transformer = transformer-based architecture.

TABLE 9. Model performance on drug interaction prediction.

Method	Year	Task Type	Macro-F1	AUROC	Macro-P	AUPR	Accuracy	Macro-R
DeepDDI [112]	2018	Multi-class	-	-	-	-	0.924	-
DDIMDL [113]	2019	Multi-class	0.759	0.998	0.847	0.921	0.885	0.718
Hou et al. [114]	2019	Multi-class	-	0.942	-	-	0.932	-
SSI-DDI [115]	2021	Multi-class	-	0.984	0.981	-	0.945	-
MDNN [116]	2021	Multi-class	0.830	0.998	0.862	0.967	0.918	0.820
MUFFIN [117]	2021	Multi-class	0.950	-	0.957	-	0.965	0.948
MUFFIN [118]	2021	Multi-label	-	0.916	0.703	-	-	-
SumGNN [119]	2021	Multi-class	0.869	-	-	-	0.927	-
SumGNN [120]	2021	Multi-label	0.876	0.949	0.934	-	0.938	0.877
META-DDIE [121]	2022	Multi-class	0.843	0.998	-	-	0.920	0.830
MDDI-SCL [122]	2022	Multi-class	0.852	0.999	0.816	-	0.908	0.839
TBPM-DDIE [123]	2022	Multi-class	0.888	0.999	0.978	0.969	0.930	0.876
DM-DDI [124]	2022	Multi-class	-	0.999	0.880	0.964	0.930	-
MDF-SA-DDI [125]	2022	Multi-class	0.888	0.985	0.870	0.974	-	-
STNN-DDI [126]	2022	Multi-class	0.929	0.983	0.970	0.979	0.953	0.972
STNN-DDI [127]	2022	Multi-label	0.982	0.997	0.974	0.996	0.828	-
GMPNN-CS [128]	2022	Multi-class	-	0.955	0.964	0.936	0.960	-
GMPNN-CS [128]	2022	Multi-label	-	-	-	-	0.982	-
LaGAT [129]	2022	Multi-class	-	0.901	0.879	-	0.967	-
DeepMDDI [130]	2022	Multi-class	-	0.983	0.909	0.792	0.897	0.837
R2-DDI [131]	2023	Multi-label	0.873	0.915	0.878	-	0.862	-
R2-DDI [131]	2023	Multi-class	-	0.936	-	-	-	-
DSN-DDI [132]	2023	Multi-class	0.969	0.995	0.994	-	0.969	-
DSN-DDI [132]	2023	Multi-label	0.988	0.999	0.999	-	0.988	-
ACDGNN [133]	2023	Multi-class	0.941	0.988	0.984	0.956	0.967	0.937
MCFF-MTDDI [134]	2023	Multi-class	0.955	0.998	0.976	0.972	0.977	0.946
MCFF-MTDDI [134]	2023	Multi-label	-	0.933	0.717	-	-	-
MolFormer [135]	2024	Multi-class	0.945	0.965	0.938	0.880	0.942	0.935
DrugGPT [136]	2024	Multi-class	0.958	0.972	0.949	0.912	0.941	0.952
TxGemma [137]	2025	Multi-class	0.962	0.981	0.958	0.935	0.952	0.960
DDI-Transformer [138]	2025	Multi-label	0.948	0.959	0.932	0.895	0.936	0.941

Macro-P = Macro-Precision, Macro-R = Macro-Recall, Macro-F1 = Macro-averaged F1-score, AUROC = Area Under Receiver Operating Characteristic curve, AUPR = Area Under Precision-Recall curve, LLM = large language model, GPT = generative pre-trained transformer.

In contrast, earlier ML models often lack the scalability required for high-dimensional biological patterns. These insights indicate that although deep learning (DL) models demonstrate superior capabilities—particularly in capturing latent patterns via agentic frameworks and LLMs—traditional machine learning (ML) approaches still hold significant value, especially in data-scarce contexts or when model transparency is essential. Ultimately, the choice between these paradigms should be guided by specific task requirements, dataset characteristics, and practical considerations such as computational cost and explainability.

Critical Appraisal:Common strategies include data augmentation, missing data imputation, and graph-based rare interaction detection methods such as GCN and deep fusion. While these strategies offer solutions for identifying rare interactions, they possess inherent risks. Specifically, the use of synthetic data augmentation may inadvertently generate artificial patterns that do not exist within authentic clinical datasets, potentially leading to misleading model outcomes. Statistical methods like semi-NMF and Bayesian networks are also applied to uncover rare patterns. Furthermore, a significant number of studies fail to validate predictions regarding rare interactions using external datasets, which raises concerns about the practical efficacy and generalizability of these approaches in real-world clinical scenarios. Additionally, performance metrics focusing on minority classes—such as recall and precision—are often absent, limiting the ability to assess the true effectiveness of rare interaction detection methods. The advantages, limitations, and potential bias sources of the reviewed AI-based DDI prediction methods are summarized in Table 10.

B. RISK OF BIAS ASSESSMENT

In this subsection, we evaluate the methodological quality of the included studies using the quality assessment criteria defined in Section III-C5 (Quality Score). Each study was scored on a 0–10 scale, and categorized according to the overall quality score.

The majority of the included studies demonstrated a low risk of bias, indicating that most recent works ensure transparency in dataset usage and evaluation metrics. However, a portion of studies (27%) exhibited moderate risk due to incomplete information about datasets or lack of external validation, while a smaller group (13%) presented high risk owing to insufficient methodological details.

The methodological quality and potential for bias in the 75 included studies were evaluated based on the predefined quality score (QS1–QS5). Table 11 provides a statistical summary of these findings. As shown in the results, (60%) of the studies (n=45) were categorized as low risk, indicating high standards of transparency in dataset selection and evaluation metrics. However, (27%) (n=20) exhibited a moderate risk of bias, often due to incomplete dataset descriptions or the absence of independent external validation to confirm the results. A smaller portion, (13%) (n=10), was identified as high risk primarily because of insufficient methodological details, which limits their reproducibility. This distribution highlights that while the majority of recent AI-based DDI research is methodologically sound, there is a critical need for enhanced transparency and rigorous validation protocols to ensure clinical reliability.

V. DISCUSSION

This study systematically reviews the machine learning and deep learning methods used for drug-drug interaction prediction. The findings suggest that machine learning and deep learning-based approaches offer significant advantages over traditional methods. However, it has also been identified that these approaches have certain limitations and aspects that require further development in the future.

Firstly, it has been observed that machine learning and deep learning models achieve high accuracy when trained with large datasets. Research indicates that methods utilizing multiple data sources can predict drug interactions with greater precision. Specifically, multi-data source fusion methods and transformer-based models, such as MDF-SA-DDI (Multi-Data Fusion and Self-Attention for Drug-Drug Interaction Prediction), can identify interactions between drug pairs more accurately. Deep learning-based approaches contribute to the development of more robust models for biological interpretation and facilitate more efficient data integration. Deep learning models typically achieve high accuracy when trained with large datasets. The findings suggest that deep learning-based approaches offer significant advantages over traditional methods. However, it is critical to acknowledge that this perceived superiority may be amplified by publication bias and the lack of direct benchmarking on identical datasets. As noted in our analysis, the absence of head-to-head comparisons utilizing consistent data sources renders claims of absolute model superiority questionable. The choice between DL and ML should therefore be dictated by specific task requirements, such as data volume and interpretability needs, rather than a generalized assumption of DL excellence. However, it is critical to acknowledge that this perceived superiority may be amplified by publication bias, favoring studies with positive DL results while under-reporting inconclusive or negative findings. This 'hype bias' often leads to the oversight of traditional ML methods, which remain robust and more interpretable alternatives for clinical applications [123], [136].

A representative example is the LDS-CNN framework integrates unified encoding with large-scale drug screening data, achieving high predictive accuracy in drug–target interaction prediction. Although primarily designed for drug–target interaction tasks, its methodological innovations highlight the potential applicability of convolutional neural network–based architectures to future drug–drug interaction prediction studies [137].

However, the success of these methods is heavily dependent on data quality. The observed disparities in data quality and the absence of standardized preprocessing protocols suggest that some reported performance gains may be artificially inflated. Furthermore, the lack of uniformity in evaluation methodologies remains a significant barrier. When studies employ different validation strategies, such as random k-fold cross-validation versus more rigorous drug-pair-out scenarios, direct comparative assessments become unreliable, complicating the identification of the most

TABLE 10. Comparative overview of AI-based methods for drug–drug interaction prediction.

Method / Approach	Advantages	Limitations	Potential Bias Sources
CNN / DNN	Learns complex patterns from raw data; good for large datasets	Requires large labeled datasets; uninterpretable models nature	Publication bias; dataset bias
GCN / Graph-Based	Captures relational structure in biomedical networks	Computationally intensive; limited interpretability	Over-representation of well-studied drugs
Transformer-Based	High accuracy on molecular data; leverages large unlabeled corpora	Performance may drop on small datasets	Benchmark dataset bias
LSTM / Sequence Models	Models temporal dynamics; good for time-series drug effects	Overfitting with limited sequences	Small-sample bias
Similarity-Based Models	Simple, interpretable; works well on small datasets	Less effective for complex interactions	Feature engineering bias
Ensemble ML Models	Robust performance; combines multiple learners	Sensitive to feature selection	Human bias in feature choice
Data Augmentation	Addresses class imbalance; increases training diversity	Risk of synthetic artifacts	Artificial data bias
Imputation (kNN, SVD, PCA)	Recovers missing values; improves dataset completeness	May distort underlying distributions	Imputation bias

CNN = convolutional neural network, DNN = deep neural network, GCN = graph convolutional network, LSTM = long short-term memory, ML = machine learning, kNN = k-nearest neighbors, SVD = singular value decomposition, PCA = principal component analysis.

TABLE 11. Methodological quality assessment and risk of bias distribution among selected studies.

Risk Level	Number of Studies	Percentage (%)
Low	45	60
Moderate	20	27
High	10	13

effective DDI prediction architectures. Incompleteness and imbalance in datasets may limit the models' generalization capacity, leading to incorrect predictions for certain drug combinations. In many cases, the limited availability of labeled pharmacological data leads to models that are prone to overfitting, performing well in internal validations but failing in real-world clinical scenarios. To ensure reliability, future research must shift away from relying solely on internal cross-validation and prioritize the use of independent, external test sets to provide a more accurate assessment of a model's generalization capability. Techniques such as data augmentation and transfer learning effectively address missing data; however, their application requires caution. The reliance on synthetic data can introduce artificial biases, and without rigorous validation on independent external datasets, the reported success in detecting rare interactions may not translate to actual clinical practice. Consequently, the lack of external validation remains a critical gap that complicates the transition of AI-based DDI tools into reliable clinical decision support systems. Despite the rigorous methodology followed in this systematic review, several limitations must be acknowledged. One of the primary limitations of this review is the exclusion of studies published in languages other than English, which may have led to the omission of significant regional research. Furthermore, the limited representation of studies from developing countries may affect the generalizability of our findings to diverse global healthcare settings. While this review aims to provide a comprehensive overview, these factors introduce a potential geographical bias. Future research should prioritize the inclusion of multilingual databases and broader regional studies to mitigate this gap and provide a more truly global perspective on AI-based DDI prediction.

In addition to these technical solutions, regulatory initiatives—such as China's recent generic drug approval reforms, which have improved transparency and public access to standardized drug datasets can help alleviate these limitations by providing more consistent and comprehensive data sources for model training. This expanded availability of high-quality, standardized data has the potential to enhance model reliability and facilitate broader applicability in clinical decision-making. In parallel, the integration of artificial intelligence into the drug development pipeline provides a transformative framework to overcome the limitations of traditional methods. Figure 8 illustrates the comparison between conventional and AI-assisted drug development processes, highlighting how improved data availability and advanced computational techniques can accelerate discovery while reducing costs [138], [139].

Another key challenge is model interpretability and transparency. Deep learning models are often viewed as "uninterpretable models," and it is not always clear which factors the model considers when predicting an interaction. This lack of understanding can make it difficult for clinicians

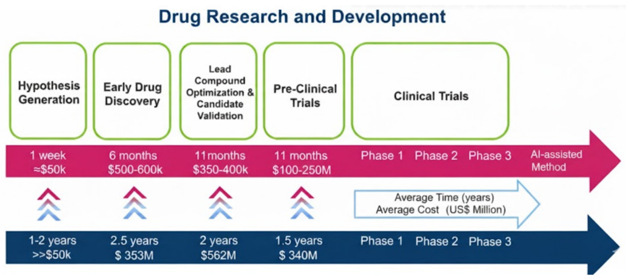

FIGURE 8. **Traditional vs. AI-assisted drug development pipelines highlighting the role of standardized data and AI integration.**

and pharmaceutical researchers to make reliable decisions based on model outputs. To overcome the 'uninterpretable models' limitation, improving the explainability of deep learning models through methods such as attention visualization or feature attribution is essential to build clinicians' trust in AI-generated outputs. This transition from reactive monitoring to a preventive, data-driven approach requires models that are not only accurate but also transparent in their reasoning. The integration of explainable artificial intelligence techniques can make the internal workings of the model more understandable, thereby enabling more reliable decision-making in clinical applications.

Although we applied strict inclusion and exclusion criteria, potential selection biases may still exist. For instance, the exclusion of non-English studies may have led to the omission of relevant findings from non-English-speaking countries. Additionally, studies relying solely on certain databases might introduce publication bias, favoring well-resourced institutions or countries. While the majority of included studies come from the United States and China, there is limited representation from developing countries. This intrinsic bias in the underlying training data suggests that the predictive accuracy of DDI models may not be uniform across different healthcare systems, particularly in low-to-middle-income regions where drug usage patterns and population genetics may differ. To overcome these constraints, future research must prioritize the development of more heterogenous and geographically diverse datasets to ensure that AI-based pharmacological tools are truly global in their clinical utility.

In light of recent developments, the usability of Large Language Models (LLMs) for drug interaction prediction and analysis has become an important research topic. Models such as Google's TxGemma have the potential to understand biomedical texts and predict drug-drug interactions. These models can open new horizons in literature review and text mining approaches, and can be especially effective in analyzing large-scale biomedical literature. Future studies should investigate how such language models can be used in drug-drug interaction prediction [140].

Additionally, current machine learning and deep learning-based approaches are typically retrospective analyses. However, proactive prediction of drug interactions requires the development of real-time data processing and updatable models. In this regard, approaches such as federated learning and continual learning show promise as solutions for predicting drug interactions more dynamically and reliably.

In conclusion, this study comprehensively evaluates the current state of machine learning and deep learning techniques used for drug-drug interaction prediction and highlights research gaps in this field. Future work should focus on developing new algorithms to enhance model accuracy, creating more comprehensive and balanced datasets, and improving the interpretability of model outputs. To improve the comparability and reproducibility of studies in this field, future research must prioritize the development of standardized datasets and more rigorous evaluation metrics. Establishing a unified benchmarking protocol that mandates consistent validation strategies (e.g., cold-start drug scenarios) and comprehensive metric reporting is essential to reliably determine the architectural superiority of emerging models. Future work should also ensure a balanced evaluation of both complex and traditional models. Strengthening evaluation standards is a critical direction to prevent the artificial inflation of performance metrics and to mitigate the effects of publication and hype biases. Ensuring that simpler, effective alternatives are not eclipsed by deep learning trends will lead to more reliable and ethically standing clinical tools. This will open the way for the more reliable and effective use of these technologies in clinical applications.

A. CLINICAL IMPLICATIONS AND PUBLIC HEALTH IMPACT

The integration of AI-driven drug–drug interaction (DDI) prediction models into clinical decision support systems (CDSS) represents a major step toward translating computational advances into clinical practice. These systems can assist clinicians at the point of care by automatically detecting and alerting potential adverse drug interactions before prescriptions are finalized. Such real-time feedback mechanisms are particularly valuable in complex therapeutic settings involving polypharmacy, where the risk of adverse events and hospital readmissions remains high. By reducing preventable medication errors, AI-based DDI tools can significantly enhance patient safety, treatment efficacy, and overall healthcare efficiency.

In addition to supporting individual clinical decisions, AI-based DDI prediction can make substantial contributions to public health surveillance and pharmacovigilance. When integrated with large-scale electronic health records (EHRs), spontaneous adverse event reporting systems, and post-marketing data, these models can continuously monitor drug use patterns and identify emerging population-level safety signals. This capability enables early detection of high-risk drug combinations, facilitates timely regulatory actions, and supports evidence-based policy decisions aimed at minimizing drug-related harm. Furthermore, real-world implementation of such systems can strengthen national pharmacovigilance infrastructures, particularly in countries where manual reporting mechanisms are limited.

From a translational perspective, ensuring clinical validation, interpretability, and regulatory compliance is critical for the safe deployment of these models. Future research should involve prospective validation using real-world clinical datasets to confirm the reliability and generalizability of AI predictions across diverse patient populations. Moreover, improving the explainability of deep learning models through methods such as attention visualization or feature attribution will be essential to build clinicians' trust in AI-generated outputs. Adherence to emerging regulatory frameworks, including the U.S. Food and Drug Administration's Good Machine Learning Practice (GMLP) principles and the European Medicines Agency's guidance on AI in medicine, will further ensure ethical, transparent, and accountable integration of these technologies into healthcare systems.

Ultimately, AI-assisted DDI prediction represents a paradigm shift from reactive drug safety monitoring toward a preventive, data-driven, and patient-centered approach to pharmacotherapy. By bridging computational innovation and clinical application, these systems hold the potential to not only improve individual treatment outcomes but also strengthen the overall safety and resilience of healthcare systems worldwide.

VI. ETHICAL CONSIDERATIONS

While AI-based drug discovery has the potential to revolutionize healthcare, the use of these technologies in clinical applications also brings with it significant ethical issues. Data privacy, in particular, is one of the most sensitive points in this area. Distributed learning approaches such as federated learning, which allow patient data to be processed without being collected in centralized systems, play an important role in protecting personal information. This method protects patient privacy while also enabling multi-center collaborations and allowing the development of more comprehensive models [141].

However, datasets used in AI systems often contain demographic biases, which can lead to the exclusion or misclassification of some patient groups. Imbalances, especially on characteristics such as age, gender, and ethnicity, can cause injustice in model performance. In order to reduce this problem, attention should be paid to the representativeness of the dataset, bias analysis should be performed in the training data, and fair modeling techniques should be applied [142].

In addition, the fact that AI models generally operate as uninterpretable models is another source of ethical concern in terms of model explainability and reliability. In clinical decision support systems, Explainable Artificial Intelligence (XAI) techniques play a crucial role in meeting ethical requirements by ensuring that decisions are transparent and accountable, thereby fostering trust among both healthcare professionals and patients. In this context, explainable Artificial Intelligence (XAI) techniques play an important role in meeting ethical requirements by ensuring that decisions are transparent and questionable [143].

VII. CONCLUSION AND FUTURE WORK

Machine learning and deep learning methods have become increasingly important in predicting drug-drug interactions. In this study, we systematically examined various machine learning and deep learning approaches used for drug interaction prediction, highlighting their advantages, challenges, and future research directions. The findings indicate that deep learning models offer higher accuracy compared to traditional methods, but challenges such as data imbalance, model interpretability, and generalization remain significant obstacles. A critical finding is that most current models are trained on datasets predominantly from high-income countries, which creates an intrinsic bias that may limit their global applicability and clinical reliability.

Future research should focus on integrating multi-modal data sources to improve prediction accuracy. The combination of molecular structures, clinical data, and electronic health records can enable models to make more comprehensive predictions. To address the aforementioned bias, future studies must prioritize the development of more geographically diverse and heterogeneous datasets to ensure that AI-based tools are effective across different healthcare systems and populations. While deep learning models offer superior accuracy, challenges such as model interpretability remain significant obstacles; therefore, the adoption of explainable AI (XAI) techniques is not merely a technical preference but a necessity to enhance the reliability and ethical standing of these models in clinical applications. Federated learning and transfer learning approaches, which preserve data privacy while enabling collaborative efforts, may provide solutions to the problem of data scarcity.

Large language models, especially models specialized for the biomedical domain such as Google's TxGemma, can be considered as a new research direction in drug-drug interaction prediction. These models can help analyze the vast biomedical literature, explore complex relationships between drug properties, and predict potential interactions. Future studies should emphasize the establishment of standardized benchmarks and more robust validation protocols, specifically necessitating the use of independent test sets to counteract overfitting and prevent the reporting of artificially inflated performance metrics. Furthermore, the further exploration of reinforcement learning and graph-based models holds great potential, particularly for predicting novel drug combinations. Future research must prioritize the validation of rare interaction predictions using diverse, real-world external datasets to confirm their practical efficacy. Additionally, developers should focus on creating data augmentation techniques that preserve the authentic biological patterns found in clinical data, ensuring that AI models are both robust and clinically trustworthy.

To improve the comparability and reproducibility of studies in this field, future research must prioritize the development of standardized datasets and evaluation metrics. Strengthening these standards is a critical direction to prevent the artificial inflation of performance metrics caused by methodological disparities and to ensure the reliable integration of AI-based drug interaction prediction into global healthcare systems. Overcoming these challenges will make a significant contribution to the more reliable and effective integration of AI-based drug interaction prediction into drug development processes and personalized medicine globally.

ACKNOWLEDGMENT

The authors acknowledge the use of artificial intelligence (AI) tools in preparing this manuscript. ChatGPT (OpenAI) was employed for grammar refinement, language translation, and generating preliminary conceptual drafts for some figures before producing the final illustrations. All AI-assisted content was reviewed, edited, and validated by them to ensure accuracy and integrity.

DECLARATIONS

- Funding
  No funding was received for conducting this study.
- Conflict of interest/Competing interests
  The authors declare that they have no competing interests.
- Ethics approval and consent to participate
  Not applicable.
- Consent for publication
  Not applicable.
- Data availability
  The data supporting the findings of this study and the search strategies are available upon request from the corresponding author.
- Materials availability
  Not applicable.
- Code availability
  Not applicable.
- Author contribution
  Zeynep Barut conceived the study, designed the review protocol, performed the literature search and data extraction, conducted the analysis/synthesis, prepared the figures and drafted the manuscript. Turgay Tugay Bilgin verified the methodology, supervised the study, contributed to the analysis and interpretation, and revised the manuscript critically for important intellectual content. All authors read and approved the final version and agree to be accountable for all aspects of the work.

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